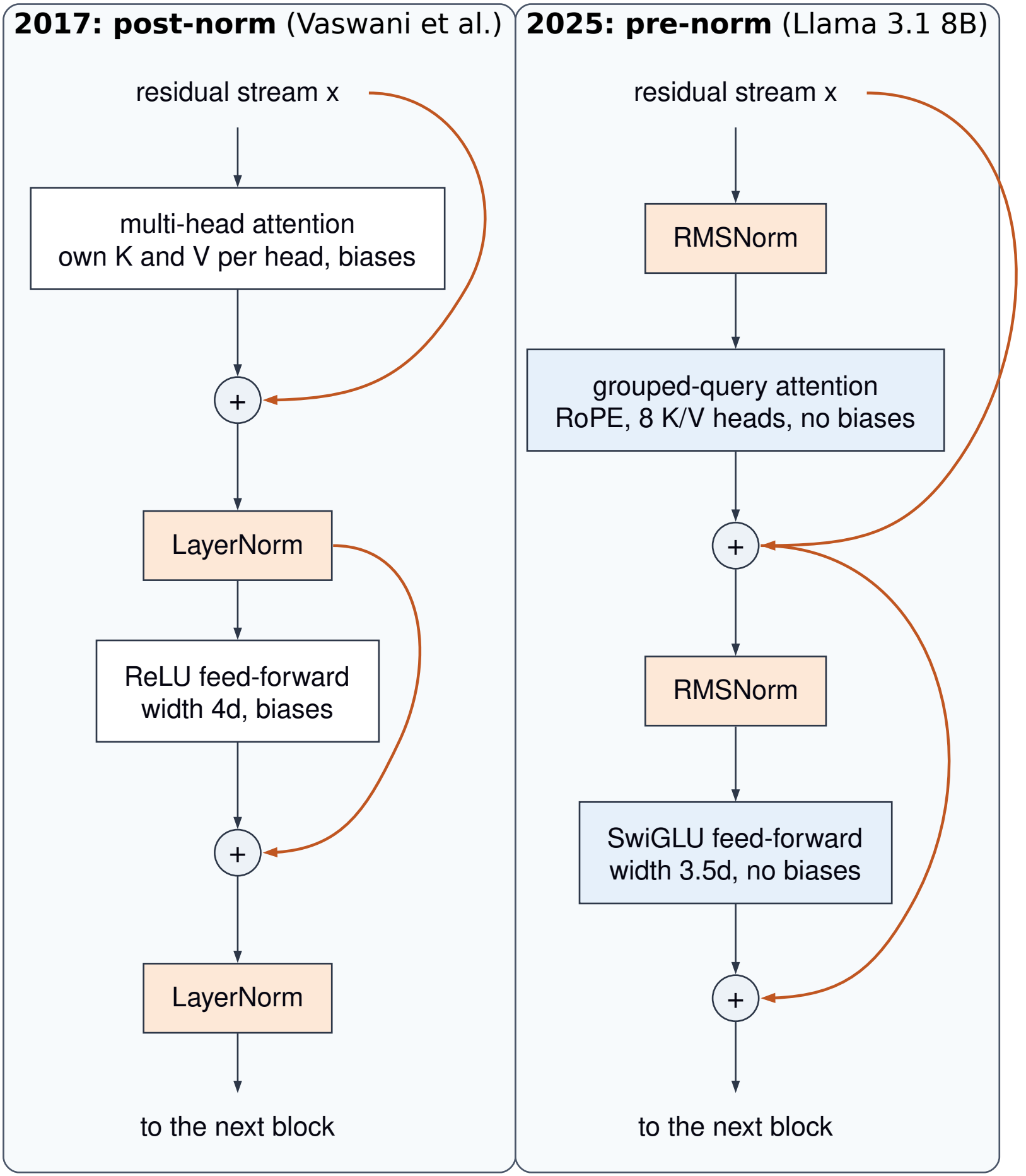
2017: post-norm (Vaswani et al.)
residual stream x
multi-head attention
own K and V per head, biases
+
LayerNorm
ReLU feed-forward
width 4d, biases
+
LayerNorm
to the next block
2025: pre-norm (Llama 3.1 8B)
residual stream x
RMSNorm
grouped-query attention
RoPE, 8 K/V heads, no biases
+
RMSNorm
SwiGLU feed-forward
width 3.5d, no biases
+
to the next block